



Review of hybrid aerial underwater vehicle: Cross-domain mobility and transitions control[☆]

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ABSTRACT

Recently, there has been growing interest in developing hybrid aerial vehicles (HAUV) capable of operating in air and water and performing continuous and uninterrupted high-quality observation and sampling. This paper provides a detailed literature review of existing HAUVs characterizing their designs, and describes the current state of the art in methodologies to enable cross-domain mobility of HAUVs. The Control system is identified as the core and crucial component to ensure repeated smooth air-water transitions for HAUV. The complete dynamic model and cross-domain motion control of HAUV under the complex sea state with wind and wave interferences have received much attention from the research community. This paper presents a review of the leading research works focusing on these technologies. In this survey paper, we introduce a systematic categorization of the HAUV platform designs and discusses their cross-domain mobility and characteristics. We also present a review of recent developments in the transition control of HAUV and the growth of related research. Finally, the paper discusses the key observations and highlights challenges that need to be addressed in developing HAUV with persistent autonomy.

1. Introduction

At present, the submersibles (Park and Choi, 2010), including autonomous underwater vehicles (AUVs), underwater gliders (UGs) as well as remotely operated vehicles (ROVs), are essential tools for the human to explore, understand and utilize the ocean. These conventional submersibles are underwater platforms that carry different exploration equipment or operational tools for underwater observation, detection, and operation tasks. These submersibles have been proven to help human beings to perform better ocean sampling and observation, underwater exploration, underwater monitoring and patrolling, underwater operations, underwater search and rescue, and extra. In addition, cutting-edge underwater observation technologies such as bionic submersibles (Costa et al., 2017; El Daoui et al., 2014; Kelasidi et al., 2017; Kruusmaa et al., 2014) or multi-vehicle hybrid operations (Conte et al., 2016) can significantly improve the efficiency of underwater propulsion

or operational efficiency, these developments have advanced the capabilities of underwater vehicles. Today, as marine science rapidly develops and human activities in the ocean continue to diversify, many emerging ocean science and engineering needs arise, including the urgent need for joint air-sea observations missions. These missions include observing ocean phenomena with high spatial and temporal variability in air and sea environments, such as boundary currents, upwelling, eddies, and fronts. As a result, scientists can obtain more prosperous and more comprehensive scientific data on these physical ocean phenomena promptly; These missions also include monitoring ocean emergencies such as red tides and oil spills from the air, detecting them underwater, and sampling the water in the area. By implementing aerial monitoring, underwater detection, and water sampling in the area of occurrence, we can observe these events in real-time and analyze and predict the development of these events simultaneously. However, the existing submersible technology can no longer meet the requirements of the

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emerging air-sea cross-domain missions, as these underwater mobile platforms cannot cross the water surface into the air for aerial observation and operations. Therefore, joint air-sea observations cannot be achieved by relying solely on existing conventional submersibles.

To provide a more convenient and efficient solution for air-sea cross-domain missions, an emerging concept, the hybrid aerial underwater vehicles (HAUVs), has significantly expanded in recent years. HAUVs are amphibious vehicles that can operate both in the air and underwater and achieve the application goal of simultaneous air, surface, and underwater exploration of a specific sea area with a single vehicle. Advanced autonomous control technology allows the operator to remotely deploy HAUVs onshore or ship deck and command it to fly autonomously to a set target area. The HAUV can also fly back to the designated location when the mission is completed for recovery. Therefore, HAUVs can provide more efficient means of implementing ocean observation with their amphibious locomotion capability, significantly reducing human intervention and saving time and manpower costs. With this unique capability, HAUVs have great potential for joint air-sea scientific research missions. Such as observing from different angles, maximizing the collection of information and data for a single mission, and improving the efficiency of marine scientific observation.

Furthermore, it can be used for maritime emergency search and rescue, using its rapid movement capability to reach the accident site in time and carry out air and underwater search and rescue in relevant areas; it can also be used for maritime patrol, target detection, and tracking, and extra. Moreover, HAUVs can be used to perform some covert missions using their unique amphibious locomotion capability. It can also accomplish cross-domain signal relay missions, to be used as a mobile relay station for signal interaction between the HAUV, base stations, and satellites by the convenience of free access to water. In conclusion, HAUV is a new type of unmanned system that significantly enriches the existing autonomous unmanned observation platforms and improves the technology level of human's cross-domain exploration activities underwater and in the air. Furthermore, it will positively affect the intelligence level of ocean observation and provide new technological support for constructing a sophisticated sea and air stereoscopic observation system.

However, in practice, HAUVs as a new platform still face many challenges and are not mature enough to accomplish the above-mentioned applications. Two of the most critical issues are (1) "how to improve the HAUV's cross-domain mobility?" and (2) "how to ensure repeated smooth air-water transitions for HAUV?".

1.1. Motivations

Since the nature of water and air are entirely different, there are many fundamental differences between the design approaches of aerial vehicles and submersibles. The new concept, HAUVs, requires a design approach that considers the different environmental constraints underwater and in the air. Unfortunately, there are currently no mature solutions worldwide that can meet the requirements of actual combined air and underwater missions. Based on the current research results, the main HAUVs are currently designed by adapting fixed-wing UAVs or multi-rotor UAVs to the underwater environment and making the power units sealed with waterproofing structures to enable them to operate underwater. Both types of HAUVs have their unique advantages and disadvantages. The fixed-wing HAUVs inherit the efficient aerodynamic characteristics of the fixed-wing UAVs and have a long-range and high flight speed. However, they are less maneuverable and flexible than the multi-rotor HAUVs, and cannot fly at low speed or hover in the air. At the same time, because of the lack of stable motion at low speed, fixed-wing HAUVs can only operate at high speed across the medium, for example, entering the water in a fast dive or exiting at high speed, or taking off and landing in a gliding mode like conventional seaplanes. Therefore, it is difficult for fixed-wing HAUVs to avoid significant fluid

resistance and impact when entering and exiting the water, requiring more stringent structural safety, dynamic performance, and handling technology during cross-domain operation.

In contrast, the multi-rotor HAUVs inherit the excellent maneuverability of multi-rotor UAVs, with stable low-speed flight, vertical take-off and landing (VTOL), and fixed-point hovering capability. The unique VTOL function is ideal for use in complex and constrained environments. More importantly, the VTOL approach to cross-domain movement ensures smooth, reliable, low-resistance, and safe access to the water, which is essential for a HAUV that needs to cross the water-air interface repeatedly for amphibious missions. However, the slow flight speed and high power consumption of the multi-rotor HAUVs make their flight range very limited and unable to adapt to the requirements of long-distance transfer capability for large-scale joint air-sea observation missions. The above analysis shows that the existing fixed-wing HAUVs and multi-rotor HAUVs have their performance shortcomings, hence unable to fully fulfill the practical application requirements of joint air-sea observation missions. In addition, current research efforts have not considered the need to find a system solution that is genuinely suitable for underwater use. Therefore, multimodal HAUV was developed to expand the underwater capability of HAUV. Multimodal HAUV evolved from the fixed-wing take-off and landing UAV. It combines the hovering ability of multi-rotor HAUV and the long air endurance feature of fixed-wing HAUV. Meanwhile, the amphibious wing can achieve underwater glide locomotion. This strategy allows multimodal HAUV to move underwater in the form of low-power gliding, and underwater endurance can be improved relative to the first two forms. However, it is not as maneuverable as multi-rotor HAUV, and its air flight endurance is not as good as that of a fixed-wing HAUV.

Another critical issue that must be adequately addressed in the field of HAUV technology is the control of HAUVs when crossing the medium. Regardless of the category of the HAUV, realistic air-sea, trans-domain application scenarios require the ability to cross the water-air interface and then move into different environmental media. In the process of crossing the medium, the HAUV will encounter very complex dynamics problems. The first and foremost is the unknown disturbance of the surrounding and time-varying change in dynamic response characteristics caused by the change of the environmental medium. Due to the considerable difference in density and viscosity between water and air, even if a vehicle is operated in the same state in both media, the vehicle's external environmental forces and the consequent dynamic response behaviours are significantly different. Therefore, when the HAUV is operating across the medium, the dynamics related to the additional mass effect, viscous damping effect, and actuator performance will have prominent time-varying characteristics, making HAUV system parameters vary with the system state during the trans-media process. In addition, the HAUV system itself is highly nonlinear, and the system characteristics that cannot be accurately modeled introduce uncertainties into the mathematical model of HAUVs. Thus, the nonlinearity, time variability, and uncertainty faced by HAUV systems make the motion control of HAUV cross-domain processes more complex than that of conventional UUVs and UAVs in a single medium. In addition to the abovementioned effects of environmental changes on the intrinsic properties of the system, HAUVs face adverse effects from environmental disturbances during the cross-domain process. Compared with any previous UUV, USV, or UAV, the environmental disturbance components encountered by HAUVs during cross-domain in natural sea conditions are more complex, and HAUV will experience the combined disturbance of unknown wind and waves. We refer to the influence of superstructure design on the transverse sway of a large vessel in cross-wind conditions, and when the HAUV is partially submerged, the different loads generated by wind and waves on the hull may cause the hull to tilt severely. If not designed properly, the body may eventually capsize, and the cross-domain movement may fail, especially when existing water. The essential technological means to deal with the above problem is to design a reasonable and reliable motion controller for

HAUVs, which can effectively and stably control the system to overcome environmental changes and disturbances during the cross-domain process.

2. Current status of HAUV

Since the 1930s (Reid, 2012), air-sea trans-domain vehicles, which can travel freely through water and air without environmental restrictions, have become one of the most imaginative creations of humankind. Unfortunately, due to the limitations of the technology level and the complexity of operating systems, no successful human-crewed trans-domain vehicles have been seen to date. In recent years, the maturity of unmanned systems technology and its widespread and successful applications in sea, land, and air scenarios have led to a renewed interest in the possibility for the development of the HAUV. The emerging HAUV technology is currently in its initial exploration phase, with many novel design solutions being proposed and validated. Existing HAUVs can be classified into various types from different perspectives. In this paper, existing forms of HAUV are grouped into two dominant designs. One is the bioinspired HAUVs, they take design inspiration from creatures in nature that can move both in the air and underwater. For example, cliff puffins, boobies, flying fish, frilled squids, dragon lice insects, and extra (Alzu'bi et al., 2018). For bioinspired HAUVs, although some research results have successfully

demonstrated the cross-domain motion capability of this novel vehicle, its technology maturity is still low, and many vital technologies still require further developments to show full potential. The other form is HAUVs developed based on the design concepts of mature UAV and UUV systems, most of this class of HAUVs can reserve good underwater or airborne locomotion capability while achieving reliable and stable cross-media motion.

2.1. Bioinspired HAUV

Bioinspired HAUVs usually fall into these three categories (as shown in Fig. 1): (1) Flapping wing, which imitates how fish, birds, and insects fly and swim by flapping wings and fins. (2) Water jet propulsion, which imitates the way squids use water jets squeezed out of their body for propulsion. (3) Plunge-dive, which imitates how sea birds, for example gannet, enter the water.

2.1.1. Flapping wing bioinspired HAUVs

Flapping wings or fins feature this type of HAUV for aerial or underwater propulsion, like fish, birds, and insects. In 2010 and 2011, a team from MIT developed a robotic flying fish and analyzed its capability of escaping water surface. The vehicle is designed to move underwater with its flapping wings and uses the wings to glide in the air, like an actual flying fish. However, the vehicle can only maneuver

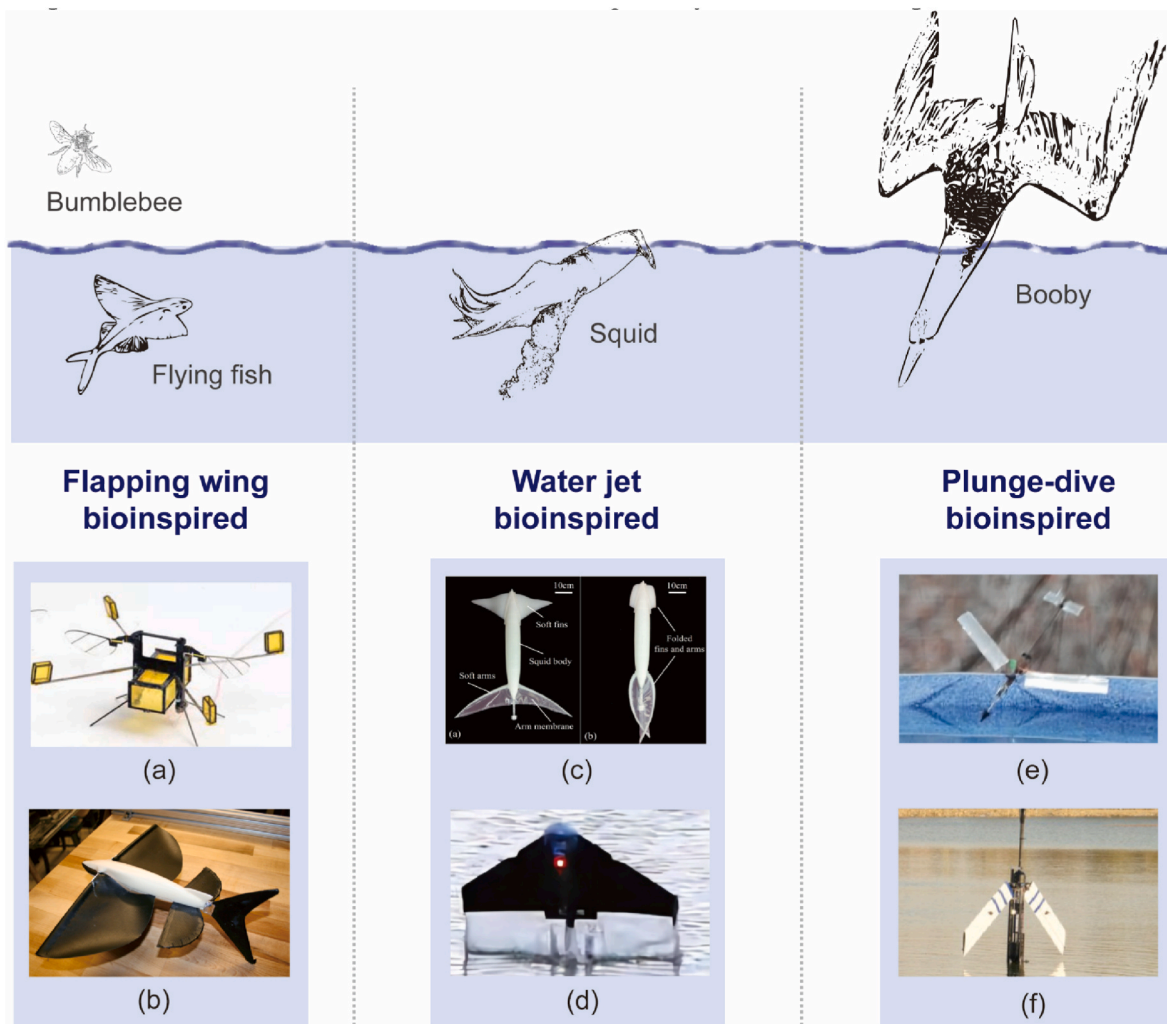


Fig. 1. Representative bioinspired HAUVs. (a) Harvard University bionic insect (Chen et al., 2017) (b) MIT biomimetic birds (Gao and Techet, 2011) (c) Biomimetic flying squid (Hou et al., 2019) (d) Chemical combustion HAUV (Zufferey et al., 2019) (e) MIT biomimetic bonito (Fabian et al., 2012) (f) HAUV with folding wings (Wang et al., 2013).

underwater and did not manage to escape the water surface. Based on the team's analysis, the requirements on the mechanical system to drive the vehicle out of water is far beyond possible (Gao and Techet, 2011). In 2017, a joint team from Harvard University and Imperial College London developed a fully functional miniature flapping wing HAUV based on a miniature flapping wing UAV RoboBee developed by Harvard University. The vehicle weighs 175 mg and has achieved stable and repeatable water entry and escape with a carefully designed control strategy. This vehicle's complete mission profile capability is proven will experiments (Chen et al., 2017).

2.1.2. Water jet bioinspired HAUV

This type of HAUV is featured by using a water jet as a propulsion medium. The use of water-jet propulsion, especially for water escape, can create consistent and predictable propulsion force, unlike propellers that vastly change propulsion characteristics due to the change of medium and become unpredictable. In 2016, a team from Imperial College London developed a HAUV prototype that uses water jet-propelled by compressed gas for water exit. The system was proven successful and later applied to the team's bioinspired HAUV, the AquaMAV (Siddall and Kovac, 2016). In 2019, the same team developed another vehicle that uses solid explosives instead of compressed gas as a pressure source (Zufferey et al., 2019). Although compressed gas and solid explosives have proven to be successful, they have one common shortcoming. The systems both use an un-reusable pressure source. As a result, the vehicle can only perform on water escape without a refuel. In 2019, Beihang University developed a flying squid with soft fins and wings supported by soft arms. The wings and fins can morph like a live organ and fulfill different aerial and underwater locomotion requirements. The aerodynamic and hydrodynamic characteristics of the vehicle are tested during experiments. The prototype adopts jet propulsion unit, which uses compressed gas in the gas cylinder to propel water jet from bionic mantle cavity (Zufferey et al., 2019).

2.1.3. Plunge-dive bioinspired HAUV

This type of HAUV is featured by the unique way of plunge-dive water entry. Some sea birds like gannet will first dive from the sky, fold their wings, and plunge into the water when pursuing prey underwater. To imitate this process, this type of bioinspired HAUVs has typically a pair of variable-sweep or foldable wings that extends in the air to generate lift for flight and folds before entering the water. A team from Beihang University in China conducted relevant research in 2011 and 2012. A computational fluid dynamics (CFD) analysis was conducted to study how impact acceleration relates to dropping height and impact angles. The team also developed a bionic gannet and conducted experiments to study the abovementioned relationship (Liang et al., 2013; Wang et al., 2013; Yang et al., 2012). The team also studied wing loading conditions of plunge-diving locomotion (Liang et al., 2014). In 2012, a team from MIT Lincoln Laboratory developed a miniature plunge-dive HAUV. The vehicle successfully achieved air-to-water transit. However, it does not have underwater locomotion or water-to-air transition capability (Fabian et al., 2012).

In 2017, a team from Imperial College London developed a plunge-dive HAUV called the AquaMAV with variable-sweep cambered thin wing and conducted a detailed analysis of water and wind tunnel performance for different wing sweep angles. The vehicle also uses water jet-propelled by compressed gas for water exit. The plunge-dive and water jet escape capability of this vehicle are proven with experiments. However, the vehicle cannot maneuver underwater yet (Siddall et al., 2017). In 2021, a Swiss Federal Institute of Technology Zürich developed a fully functional plunge-dive HAUV, the Dipper. It has a pair of actively controlled variable-sweep wings for flying. The wings can be swept back 90° for plunge-dive water entry. The vehicle uses different propellers for aerial and underwater propulsion. The vehicle is proven to have full mission profile capability (Rockenbauer et al., 2021).

We have summarized several animals and robots with aerial-aquatic

capabilities and identify the key advantages and limitations of the current bioinspired HAUVs. While these bioinspired HAUVs has a distinct set of locomotion modes, the vehicles' sizes are usually small, and the load and motion capabilities are minimal. Natural disturbances of ocean environments can be a significant problem for these small scale bioinspired HAUVs. Water waves and spray will restrict the minimum flying height, create flight perturbations and cause the vehicle to crash. Therefore, future efforts should be explored to design the bioinspired HAUVs inherently robust with intelligence transition controller.

2.2. UAV and UUV based HAUV

HAUVs significantly improve upon current UAV and UUV technologies, often involving integrating these two types of vehicles. Based on the existing research of HAUVs, this paper classifies them into three categories according to the form of the aerial system, namely fixed-wing HAUV, multi-rotor HAUV, and multimodal HAUV. These HAUVs are generally designed to perform various missions and thus use different actuators and propulsion systems. Typical missions and transition schemes compatible with the specific type of HAUVs are illustrated in Fig. 2. Generally, the fixed-wing HAUV has the characteristics of long-range flight, and the multi-rotor HAUV has the characteristics of strong maneuverability. Finally, the multimodal HAUV has the characteristics of long underwater endurance. Detailed analysis is expanded below.

2.2.1. Fixed-wing HAUV

Like fixed-wing UAVs, fixed-wing HAUVs use the lift provided by the wings to lift the vehicle in the air to achieve high-speed HAUV flight with less power consumption. Nowadays, many scholars have carried out the design and development of fixed-wing HAUV prototypes (e.g., Fig. 3).

The US Office of Naval Research proposed a fixed-wing HAUV deployed by airdrop and developed the principle prototype Test Sub II. The Test Sub II rapidly approaches the target water by gliding after airdrop and then splashing to complete the underwater mission (Edwards et al., 2017; Young, 2014), but this series of HAUVs cannot be recovered for reuse. Johns Hopkins University developed a simple triangular plate-like fixed-wing HAUV made of carbon fiber panels and foam and conducted tests focusing on tilt take-off operations (Moore et al., 2018). The test results show that the tilt-out process of the fixed-wing HAUV has strict requirements on the attitude of the vehicle and that if the attitude and speed during the take-off cannot be adequately matched, the vehicle will not be able to generate enough lift and eventually, the vehicle will crash back into the water. North Carolina State University has announced a fixed-wing HAUV. The EagleRay (Weisler et al., 2018), which is built by waterproofing a fixed-wing UAV, the critical feature is the design of an open structure that allows water to enter and exit the wing freely to reduce the excessive buoyancy of the wide wing cavity in water (Stewart et al., 2020). The EagleRay is capable of airborne flight, surface navigation, and small-depth underwater locomotion, but it is less efficient to operate underwater with an aerial propeller.

In a subsequent improvement, the research team added a stern-mounted water plunger to the EagleRay to improve the efficiency of underwater motion (Stewart et al., 2018). The RMIT researchers designed the principle prototype of the HAUV with foldable wings, considering that the broad wings of a fixed-wing HAUV would create significant drag during trans-media and even be damaged during water entry by hitting the water surface at high speed (Guo, 2019; Guo et al., 2019). In addition, the prototype is equipped with a propeller with foldable blades and a water jet take-off device using compressed gas as the working medium. The team has performed complete simulation and wind tunnel tests of this HAUV prototype in both modes (fixed-wing mode with fully expanded wings and wingless mode with fully retracted wings), but the prototype has not yet been tested in actual cross-domain

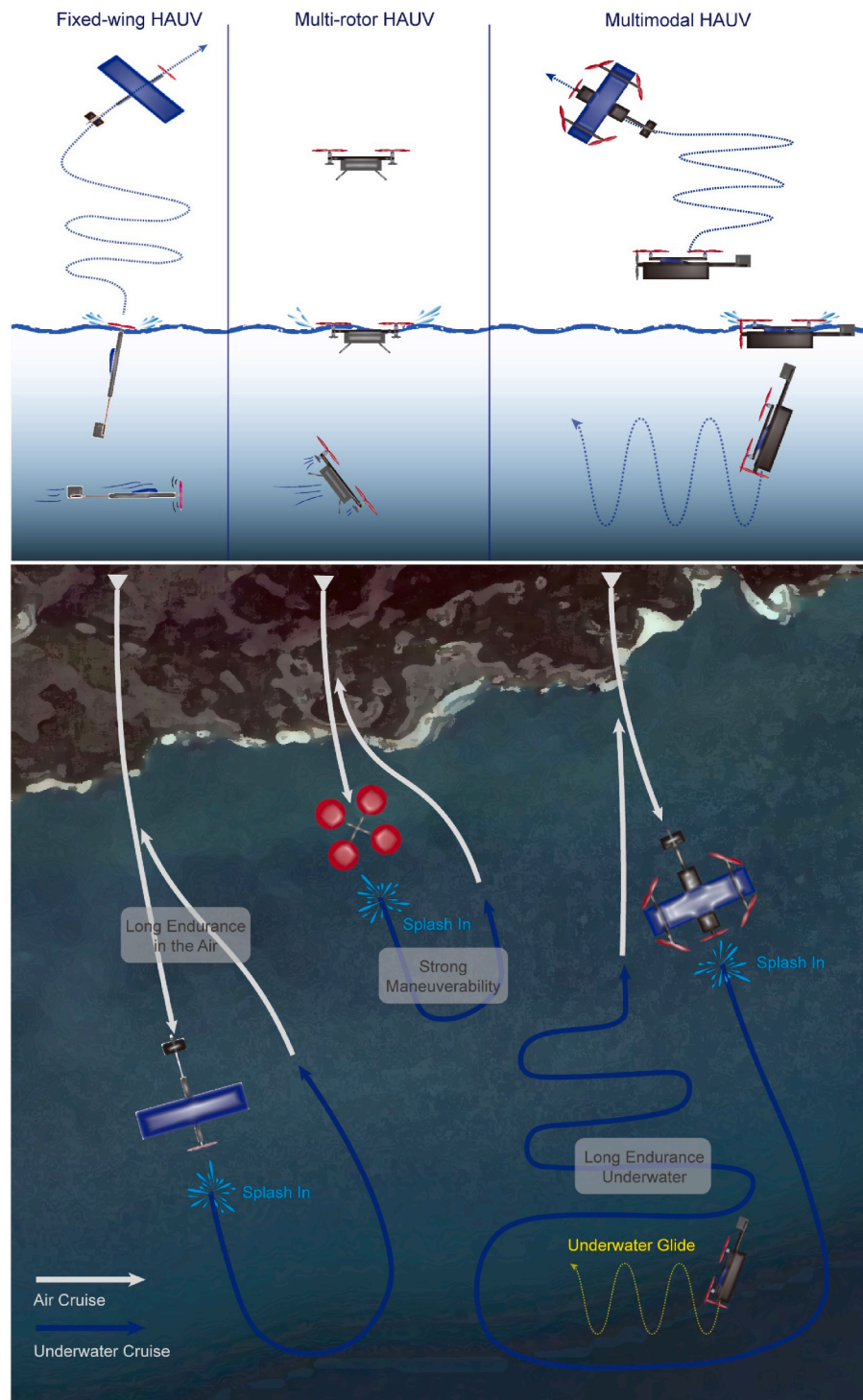


Fig. 2. Typical missions and transition schemes of UAV and UUV based HAUV.

motion. The same design has been applied to a small HAUV announced by Imperial College, the AquaMAV (Siddall and Kovač, 2014). The AquaMAV has a water jet take-off mechanism and retractable wings that can be folded to reduce impact when it dives into the water. After that, the high-pressure gas can quickly squeeze out the water to achieve jet take-off. The wings open again when the body is entirely out of the water, and the aircraft flies in fixed-wing mode. In 2019, the team enhanced the jet take-off by using chemical combustion methods instead of high-pressure gas to provide more powerful water-jet power (Zufferey et al., 2019). However, neither of the two HAUVs developed by the team

is currently capable of underwater movement, and neither can take off repeatedly due to both using a non-reusable jet take-off device. Researchers at Beihang University have developed a prototype HAUV with folding wings and ballast water tanks (Yao et al., 2014). The prototype has completed lake tests to verify basic underwater diving, surface take-off, airborne flight, and landing functions. However, the “flying fish” used the surface glide take-off method, requiring a long take-off distance and can easily be affected by surface disturbances. The group has also developed a prototype HAUV with folding wings and conducted test flights to verify it (Wang et al., 2013). The prototype was equipped

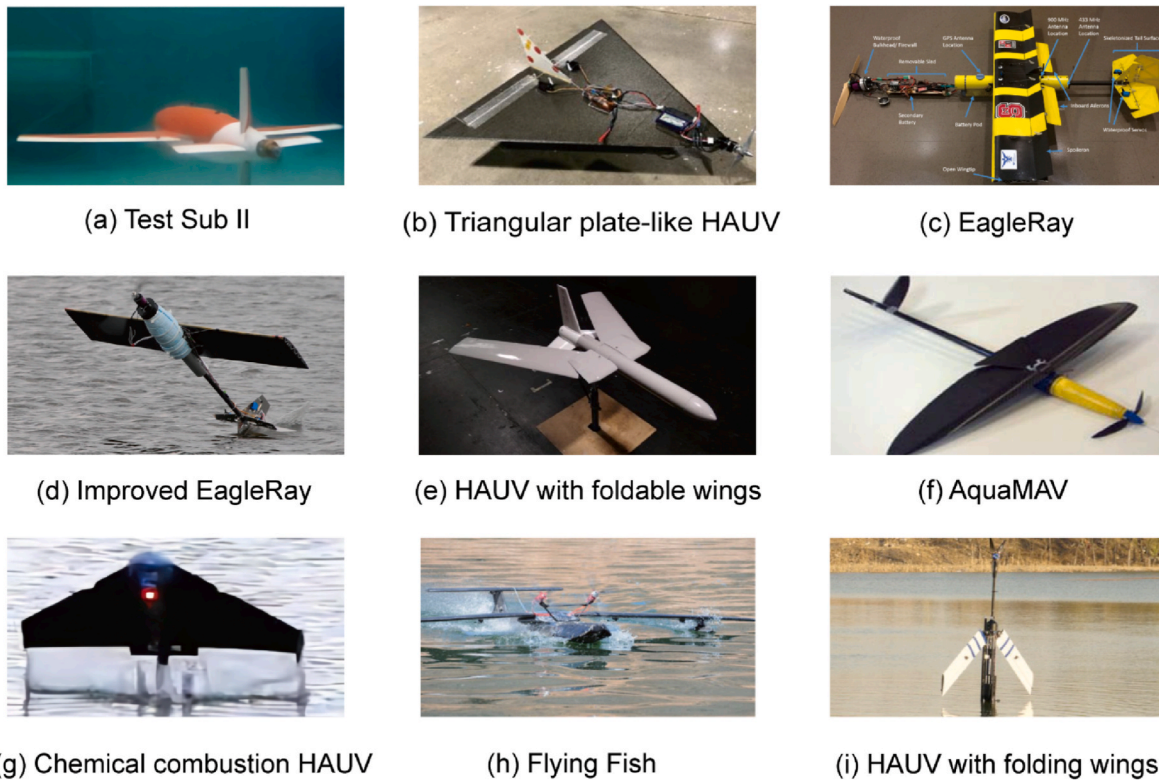


Fig. 3. Representative fixed-wing HAUVs. (a) Test Sub II (Edwards et al., 2017) (b) Triangular plate-like HAUV (Moore et al., 2018) (c) EagleRay (Weisler et al., 2018) (d) Improved EagleRay (Stewart et al., 2018) (e) HAUV with foldable wings (Guo, 2019) (f) AquaMAV (Siddall et al., 2017) (g) Chemical combustion HAUV (Zufferey et al., 2019) (h) Flying Fish (Yao et al., 2014) (i) HAUV with folding wings (Wang et al., 2013).

with airbags to assist in adjusting the attitude of the airframe on the water surface.

2.2.2. Multi-rotor HAUV

The multi-rotor HAUV adopts the basic structural framework of multi-rotor UAV, and the propulsion system composed of multiple rotors directly provide the lift and torque required for the system to fly. In addition, the unique hovering and VTOL capabilities make it easier to conduct flight, dive, and cross-domain tests on multi-rotor HAUVs. Therefore, in the past five years, multi-rotor HAUVs with significant advantages in flexibility and stability is gradually winning more researchers' favor. Rutgers University has released a series of Naviator dual-layer quadrotor HAUV prototypes (Maia, 2018; Maia et al., 2015, 2017). The Naviator uses a double-layered propulsion system for air, underwater, and vertical cross-domain locomotion through the surface. The Naviator's amphibious capability has been well demonstrated in pool tests. The team has also performed a quantitative analysis of the design and optimization of the Naviator's essential structures and equipment (Villegas et al., 2017). The main reason for adopting a power system with a two-layer multi-rotor layout is to enable seamless and

coherent movement in and out of the water. This design solution was also adopted by a joint research team from the Federal University of Minas Gerais, Federal University of São João del Rey, and Federal University of Rio Grande, Brazil, to design a two-layer quadrotor HAUV, the HyDrone (Horn et al., 2019; Rosa et al., 2018). Except that the propellers in the lower layer of the HyDrone were replaced with special propellers able to improve underwater propulsion efficiency. The prototype is still under development and has not yet been tested. A prototype HAUV with the same double-layer quadrotor structure has also been developed by the Air Force Engineering University in China (Chen et al., 2020; Ma et al., 2018), mainly for testing control algorithms. Fig. 4 illustrates the abovementioned multi-rotor HAUV system with a double-layer thruster layout.

In addition to using a two-layer propulsion layout, some scholars are also working on a single-layer multi-rotor HAUV (e.g., Fig. 5). For example, researchers at the University of California, Berkeley, have built a miniature quadrotor HUAUV with a quadrotor propulsion system that is also used to provide thrust underwater (Zha et al., 2019). The miniature HAUV has been successfully tested in a pool for active acceleration up from underwater and smoothly breaking the surface for take-off.

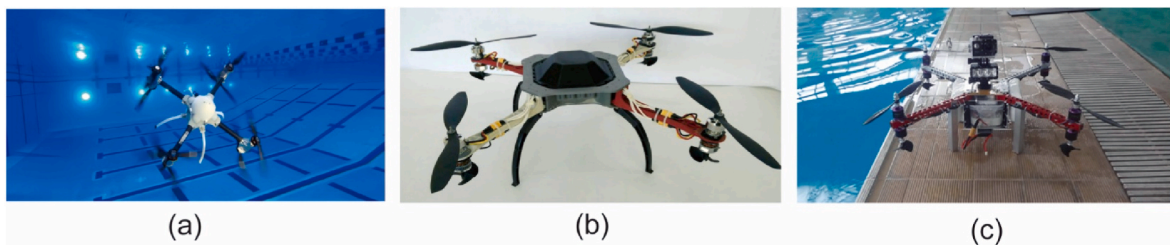


Fig. 4. Two-layered multi-rotor HAUVs. (a) Naviator 5 (Maia et al., 2017) (b) HyDrone (Horn et al., 2019) (c) Air Force Engineering University HAUV (Chen et al., 2020).

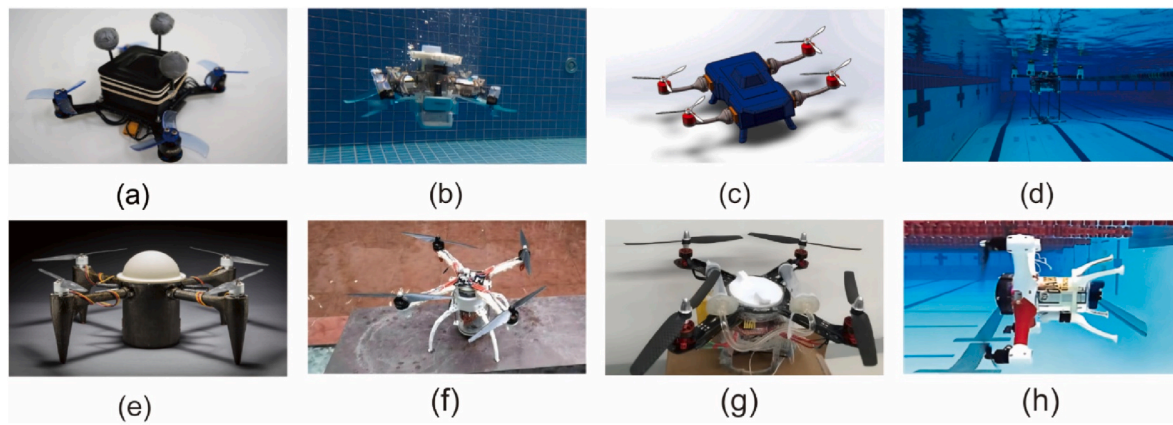


Fig. 5. Single-layered multirotor HAUVs. (a) Miniature quadrotor HUAUV (Zha et al., 2019) (b) Tilt-quadrotor HAUUV (Tan and Chen, 2020) (c) Dalian Maritime University (Chen et al., 2019) (d) GTQ-Cormorant (Bershadsky et al., 2016) (e) CRACUNS (Staff, 2016) (f) Shanghai Maritime University (Chen et al., 2021) (g) Loon Copter I (Alzu'bi et al., 2015) (h) Loon Copter II (Alzu'bi et al., 2018).

However, the rest of the maneuverability tests have not yet been conducted. Recently, researchers at the National University of Singapore have developed a novel small tilt-quadrotor HAUUV prototype. Instead of having four propellers that are fixed, this HAUUV has propellers able to change direction with a tilting mechanism. When the airframe (Tan and Chen, 2019, 2020) is in the air, the tilting mechanism turns the four rotors into a standard four-rotor UAV in a vertically upward position, while when the airframe is in the water, the four rotors can be adjusted to a horizontal or downward position and become a vector propulsion system in the water, a structure that enhances the system's maneuverability underwater. The prototype has undergone several iterations of updates and has completed tests of its amphibious movement capabilities in a pool. The Dalian Maritime University in China has also proposed a quadrotor HAUUV concept with a tilt-rotor-like structure and conducted theoretical research on its motion mechanism and characteristics (Chen et al., 2019), but no test has been carried out yet.

Continuous water exit is more difficult for single-layer multi-rotor HAUUV systems, as the rotor system needs to deal very sensitively with sudden changes in thrust characteristics as it crosses the water surface. Therefore, if there is no requirement of coherent movement of the vehicle across the surface, a buoyancy regulation scheme to help the vehicle maintain an excellent floating state (especially during water exit) until the propeller establishes a stable thrust is a good option. Georgia Tech has released a quadrotor HAUUV principle prototype, the GTQ-Cormorant. The GTQ-Cormorant can perform simple vertical snorkeling in the water by changing its net buoyancy with a buoyancy adjustment device (Bershadsky et al., 2016). The submerged function and the surface VTOL function of this prototype have been verified in pool tests. Researchers at the Johns Hopkins University Applied Physics Laboratory have developed CRACUNS (Staff, 2016), a quadrotor HAUUV prototype with a specially designed corrosion-resistant housing made of carbon fiber, which can remain submerged for long periods in shallow water and rise to the surface when needed. A quadrotor HAUUV prototype has also been developed at Shanghai Maritime University in China. It is currently being used to test cross-domain motion control of a multi-rotor HAUUV. The full amphibious motion capability of the prototype has not yet been reported (Chen et al., 2021). A quadrotor HAUUV equipped with a unique buoyancy adjustment mechanism, the Loon Copter, has been developed at Oakland University (Alzu'bi et al., 2015; Alzu'bi et al., 2018). The Loon Copter alters its weight through a pump-and-drain mechanism. When it needs to rise to the surface for take-off, the system completely drains water from the ballast tanks so that the airframe floats vertically and squarely above the surface. When the airframe needs to dive into the water for movement, it pumps water from the environment into the internal ballast tanks. The unique feature of the Loon Copter is that when the ballast tank is filled with water, the vehicle

will be tilted 90°, pointing the aerial propellers in horizontal directions to provide thrust and controls for underwater locomotion, greatly enhancing underwater manoeuvrability. The Loon Copter's complete amphibious motion capability has been successfully demonstrated in pool tests.

2.2.3. Multimodal HAUUV

Since 2017, Shanghai Jiao Tong University (SJTU) has carried out the theoretical exploration and prototype development regarding designing a new type of HAUUVs. Inspired by the traditional UG and existing HAUUV designs, SJTU proposed a new concept of multimodal hybrid aerial underwater vehicle (MHAUV) capable of vertical flight, horizontal flight, underwater gliding, and stable crossing of the water-air medium interface. The new concept of Multimodal Hybrid Aerial Underwater Vehicle (MHAUV) further enhanced the amphibious capability of the existing HAUUV system in the following aspects: (1) integrating the advantages of the existing fixed-wing HAUUV and multi-rotor HAUUV in terms of their respective flight performance and realizing the innovative design of the system that integrates high maneuverability, considerable take-off weight, high speed, and low energy consumption flight performance; (2) providing the HAUUV with the ability to glide underwater, and improving the design of the system to improve the performance of the HAUUV. The unique design concept of the MHAUV is based on the technical requirements of the actual long-range air-sea joint mission scenarios to improve the existing HAUUVs comprehensively. Therefore, the technical requirements of real-life guide the unique design concept of the MHAUV, wide-area combined air and sea mission scenarios, and aim to enhance the amphibious capabilities of existing HAUUVs. Specifically, the MHAUV adopts the UG motion mode when moving underwater and adjusts the net buoyancy and attitude of the vehicle in the water through an internal mechanism, thus achieving low energy consumption and long duration underwater gliding motion. In the air, the MHAUV can adopt two kinds of motion modes, vertical flight of multi-rotor UAV or horizontal flight of fixed-wing UAV and provide the corresponding power and lift for the vehicle in different flight modes through the coordination of rotor propulsion system and fixed-wing, to achieve hovering, VTOL or high-speed flight.

SJTU successfully developed the principle prototype of a multi-rotor HAUUV system with vertical underwater locomotion capability, the Nezha I, shown in Fig. 6 (a), in 2017. Then in 2018, SJTU successfully developed the first MHAUV prototype, Nezha II, which combines the capability of horizontal flight, vertical flight, and underwater gliding, shown in Fig. 6 (b) (Lu et al., 2018; Wei et al., 2022, 2019). At present, the "Nezha" series has completed the third iteration of the "Nezha III (Tail-sitter)" prototype, shown in Fig. 6 (c) (Lu et al., 2021b) and "Nezha III (VTOL)" prototype shown in Fig. 6 (d) (Hu et al., 2022; Lyu et al.,

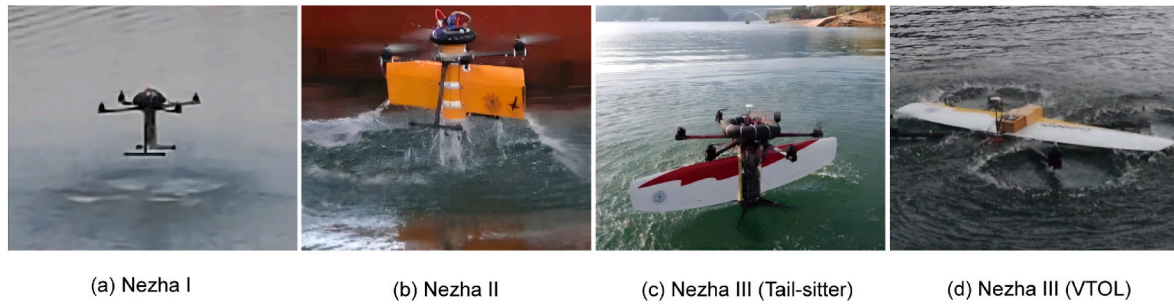


Fig. 6. The HAUV, Nezha, developed by Shanghai Jiao Tong University.

2022), it is referred the interested reader to their work for more details. Further refinement and optimization work are underway to enable MHAUV technology to meet the needs of continuous cross-domain observation.

Based on the above analysis of research progress, the performance differences between the three mainstream HAUVs (shown in Fig. 7) can be compared: (1) most of the existing HAUV systems are still initialized from the mature fixed-wing or multi-rotor UAV design, which have their advantages and disadvantages. However, all the developed fixed-wing and multi-rotor HAUV systems have minimal dive capability, which is not suitable for application in actual sea-air cross-domain observation missions, so there is still a need to improve these kinds of HAUVs further. (2) the performance advantage of fixed-wing HAUV lies in the capability of high-speed fly in the air. However, it is difficult to reduce the speed to cross the water smoothly during the cross-domain process, and its cross-domain process is more difficult to control and more susceptible to failure by wind and wave interference compared with the VTOL approach of multi-rotor HAUV. Therefore, if the advantages of the two HAUV forms can be organically integrated, the repeatability and stable maneuverability of the cross-domain process can be ensured. In contrast,

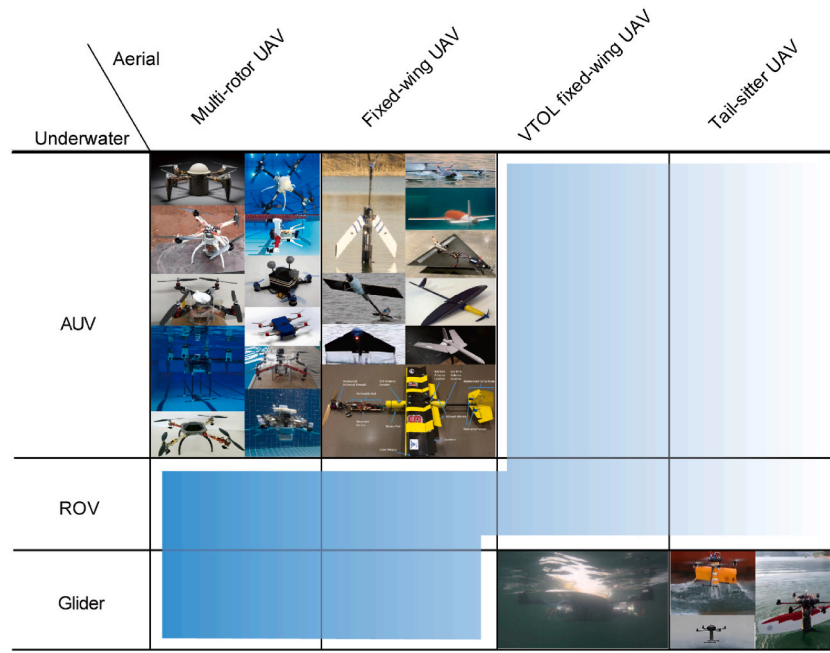
the HAUV still has the characteristics of high-speed flight, and such a HAUV system can also better meet the requirements of practical applications at sea. (3) The multimodal HAUV enhances underwater endurance and operating depth to some extent. It can ensure the repeatability and stable maneuverability of the HAUV during cross-media process while still allowing the HAUV to fly at high speed, such HAUV system can better meet the requirements of practical applications at sea.

We evaluate the existing aerial and underwater platforms in order to identify an appropriate scheme for building a HAUV. Fig. 8 (a) summarizes the configurations of the existing HAUVs, and the gaps indicate the possible forms of new configurations of the HAUVs in the future. For example, HAUV hybrid with ROV characteristic does not yet exist, this combination can achieve better underwater maneuverability. By investigating the existing HAUV, we illustrated each data-available HAUV in the Gabrielli-Von Karman graph, in Fig. 8 (b). As can be observed, existing HAUVs have very varied maximum speed and efficiency for aerial performance, due to their different form of the aerial system. It can be also observed the multimodal HAUV is more efficient for underwater operation.

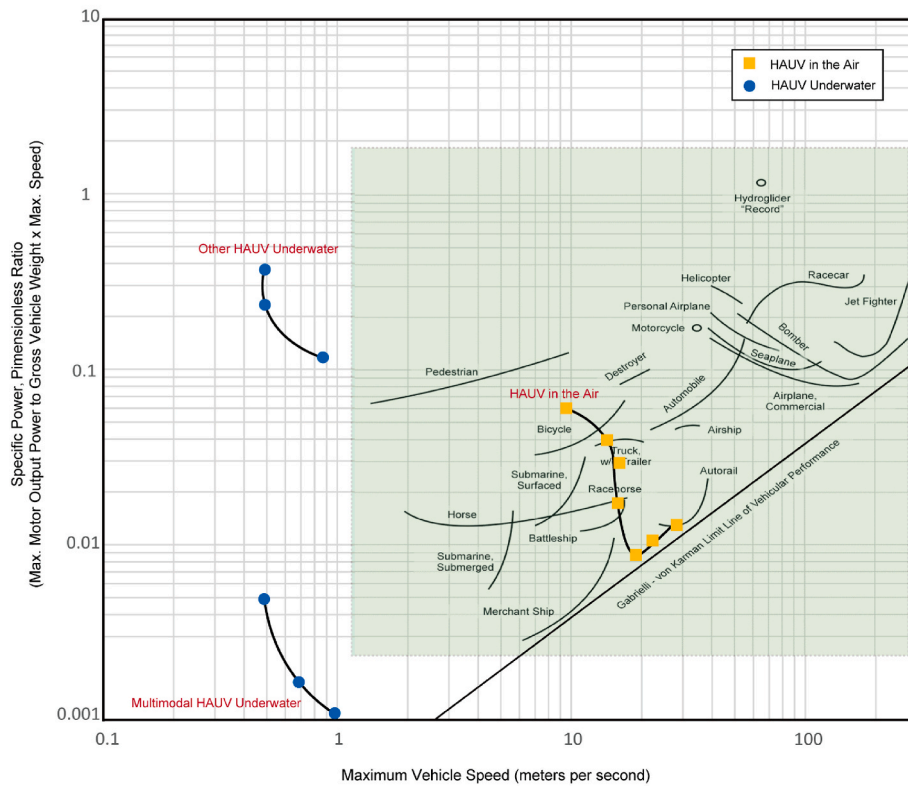
Airframe category	Fixed-wing HAUV			Multi-rotor HAUV			Multimodal HAUV		
Prototypes									
Institution	North Carolina State University	Imperial College	Beihang University	Rutgers University	University of Auckland	Shanghai Maritime University	Shanghai Jiao Tong University	Shanghai Jiao Tong University	Shanghai Jiao Tong University
Weight	5.7 kg	0.2 kg	12 kg	3.85 kg	2.7 kg	2.1 kg	4 kg	18 kg	14.68 kg
Endurance in the air	7.4 min	14 min	N/A	N/A	10.5 min	N/A	8 min	10 min	7 min
Endurance underwater	18 h	N/A	N/A	N/A	0.37 h	N/A	20 h	24 h	24.4 h
Velocity in the air	20 m/s	10 m/s	23.6 m/s	N/A	N/A	N/A	17 m/s	16.7 m/s	20 m/s
Velocity underwater	0.89 m/s	N/A	0.5 m/s	Short	0.5 m/s	N/A	0.5 m/s	1 m/s	0.7 m/s
Ability to dive	Meters	Meters	Meters	Meters	Meters	Meters	Tens of meters	Tens of meters	Tens of meters
Water escape	High speed out of the water	High pressure jet water	Surface skidded off	Continuous vertical escape	Float - Vertical take-off	Vertical launching	Vertical launching	Vertical launching	Vertical launching
Entry way	Splash/glide	Dive splash	Skidding landing	Continuous vertical	Vertical descent-Float	Vertical landing	Vertical landing	Vertical landing	Vertical landing



Fig. 7. The performance differences between the three mainstream HAUVs.



(a)



(b)

Fig. 8. (a) Attribution of existing HAUVs. The blue part represents the HAUVs vacancy for this hybrid attribute. (b) The Gabrielli-Von Karman graph of each data-available HAUV. The green part is form Radtke (Radtke, 2008). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3. Cross-domain motion control for HAUV

Stable and smooth cross-media motion control is one of the core technologies necessary to HAUV systems. In 2017, researchers at the Air Force Engineering University carried out more comprehensive modeling

and numerical analysis specifically on the variation of the hydrodynamics during water-entry and water-exit for a HAUV with elongated bodies and folding wings, mainly considering the buoyancy factors, additional mass effects, and damping characteristics that change significantly during the trans-media process (Hu et al., 2017; Qi et al.,

2017). The mathematical model was thoroughly compared and justified with CFD simulations (Zhang et al., 2018) and experimental results (Ma et al., 2020). In 2018, researchers at the Johns Hopkins University Applied Physics Laboratory modeled the triangular plate-shaped fixed-wing HAUV they had developed, taking into account the buoyancy force, additional mass effect, and damping characteristics that change during the water-exit process (Moore et al., 2018). In 2019, the same team used this model to develop a state feedback controller with a switching law for the water exit motion. Again, pool tests were conducted, but the results show that the success rate of the water exit was not ideal (8 successful take-offs out of 20 water exit tests) (Moore, 2019).

In contrast, the current research results for the cross-media motion control problem of multi-rotor HAUVs are more abundant, and one of the most important reasons is that the multi-rotor HAUV cannot perform open-loop flight operations which purely depend on manual controls, like most fixed-wing HAUVs. Instead, it must rely on a closed-loop controller to achieve stable flight. Researchers at Oakland University designed a classic proportional-derivative (PD) flight controller for the Loon Copter and successfully achieved repeatable take-offs and landings over the water surface (Alzu'bi et al., 2018). In 2014, a joint team of researchers from three Brazilian universities, Universidade Federal de São João del-Rei, Universidade Federal do Rio Grande and Universidade Federal de Minas Gerais, conducted modeling and numerical simulation research on the quadrotor-like HAUV. They added the considerations about the changing buoyancy forces, inertial and viscous hydrodynamics, and the propulsion characteristics to the kinematic model of the quadrotor UAV to mathematically generalize the main force characteristics of the quadrotor HAUV before and after the transition process (Drews et al., 2014). In their initial study, numerical simulations of the proposed system model were performed to verify the validity of the modeling approach, mainly using simple gain scheduling PD control (GSPID).

Subsequently, the team further considered the effects of different environments on the parameter variations of the vehicle's model, then linearized the model in water and air as two deterministic continuous systems respectively. This LMI-based method of H_2 robust switching control was based on the segmented Lyapunov function method to ensure the stable motion of the HAUV in different environments (Neto et al., 2015). In 2020, the team also designed a classical proportional-integral-derivative (PID) controller for its newly designed HAUV, the HyDrone, and verified the effect of cross-media motion control through simulation tests (Horn et al., 2020). The researchers at Rutgers University designed a primary GSPID controller for their HAUV, the Naviator, which switches between two sets of PID parameters based on sensor readings which reflects the vehicle's status during the water entry and water exit process. The scheme was validated by simulation and pool tests (Maia et al., 2017). The team then rederived the motion model of Naviator through the quaternion method. The GSPID control strategy succeeded in simulation (Mercado et al., 2017) and pool tests (Mercado et al., 2017) for both maneuverability operations (flip 360°) and cross-media motions. In 2018, the team proposed follow-up research that applied a hybrid control strategy to their HAUV. The approach considered the Naviator moving in the dual-domain environment as a hybrid system (a discrete step process connecting two separate continuous nonlinear systems), and then designed a composite trajectory tracking controller with a switching law based on the hybrid system theory, in which the position loop and attitude loop adopted a feedback linearization controller and a PID controller respectively (Ravell et al., 2018). This method has been validated by simulation.

In recent years, cross-media motion control of multi-rotor HAUV has become a new focus of research. Researchers at the Air Force Engineering University have conducted more research on the cross-media motion control problem of a dual-layer quadrotor HAUV. In 2016, the team addressed the propeller ground effect problem of a multirotor HAUV near water, proposed a self-anti-disturbance control strategy to achieve attitude stability control, and conducted experimental tests.

However, the study only considered the aerial flight phase near the water surface but did not involve the complete trans-media motion process (Qi et al., 2016). Attempting to use more advanced algorithms to realize the uncertainty of the system model and unknown real-time estimation compensation of time-varying environmental effects which can improve the robustness of the system against environmental changes in the cross-media process, in 2018, the team designed the adaptive sliding mode control strategy (AC) with real-time rectification parameters for cross-media motion of multi-rotor HAUVs. (Ma et al., 2018). In 2019, researchers at Dalian Maritime University designed a classical sliding mode controller (SMC) for the cross-media motion control problem of their proposed new concept, a tilting quadrotor HAUV, and achieved stable control of altitude and attitude in simulation tests of the cross-media process (Chen et al., 2019). In 2021, researchers at Shanghai Maritime University designed a "Fuzzy Proportional-Integral-Integrative" (Fuzzy P + ID) attitude controller for a single-layer quadrotor HAUV prototype to maintain the attitude stability of the system during the trans-media motion. Since the Fuzzy P + ID controller has the same structure as the classical PID controller, it is easy to implement in hardware, and the effectiveness of the method has been verified in the pool tests of the prototype (Chen et al., 2021).

Recently, researchers from Shanghai Jiao Tong University have conducted a detailed modeling study on the motion model of multi-rotor HAUV in water, air, and cross-media processes and introduced the effects of wind, wave, and current into the system model for the first time. An adaptive dynamic surface control (ADSC) strategy was applied for the HAUV to track the desired altitude and attitude during the transition. The ADSC scheme can actively adjust the control outputs according to the change of media, which improves the environmental adaptability of the whole control system and thus enables stable air-water transition and good altitude and attitude tracking performance (Lu et al., 2020). Subsequently, a nonlinear disturbance observer (NDO) based on the dynamic surface controller was designed to track the trajectory of the proposed HAUV under complex sea states with wind and wave interferences (Lu et al., 2021). It was testified that the control scheme could accurately estimate the total disturbance in real-time during the transition stage, which improves the system's robustness against the varying uncertainties and disturbances. It thus enabled smooth vertical take-off and landing (VTOL) at the surface and good trajectory tracking performance even in the presence of wind and wave.

The contents of the above research work are summarized in Fig. 9. It can be seen that: (1) Exploring the modeling methods of HAUV and conducting relevant modeling studies are the basis for the research on cross-media motion control of HAUV systems. Especially for developing a new HAUV system, establishing a reliable mathematical model is essential to carry out the subsequent controller design work; (2) In multirotor HAUV prototype platforms that have been successfully tested so far, almost all of them still adopt the classical PID control strategy with simple structure and manage to achieve good effectiveness. However, PID controllers cannot proactively adjust control output characteristics when facing environmental changes. Sudden changes introduced by the application of switching control methods can easily cause jitters. These problems can be effectively solved by using adaptive algorithms to assist the design of a global continuous nonlinear controller; (3) HAUVs operating in actual sea conditions will inevitably encounter wind and wave interference. Dynamic surface control method with disturbance observer has the ability to actively estimate and compensate for disturbances, effectively solving the problem of stable surface take-off and landing of the system in complex sea conditions.

4. Open problems and research challenges

Up to now, the emerging technology of amphibious vehicles in the air and sea is still in its infancy, and the research in related fields is in its initial stage, so it is one of the most critical tasks nowadays to actively explore the appropriate and effective forms and design methods of

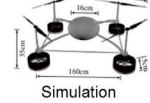
Property Controllers	Continuity	Nonlinearity	Robustness	Model dependency	Computational complexity	Implementation	Papers	Examples
PID	N	Y	N	Y	Y	Y	(Alzu'bi et al., 2018; Horn et al., 2020)	 Experiment
GS-PID	N	Y	N	Y	Y	Y	(Maia et al., 2017; Mercado et al., 2017)	 Experiment
Fuzzy P+ID	N	Y	Y	Y	N	Y	(Chen et al., 2021)	 Experiment
AC	Y	Y	Y	N	N	N	(Ma et al., 2018)	 Simulation
SMC	Y	Y	Y	N	N	N	(Chen et al., 2019)	 Simulation
H ₂	Y	N	Y	N	N	N	(Neto et al., 2015)	 Simulation
ADSC	Y	Y	Y	N	N	N	(Lu et al., 2020)	 Simulation

Fig. 9. Controller performance comparison.

amphibious vehicles. As can be observed from systematic categorization of the existing HAUV designs, it seems the multimodal HAUV more potential to achieve the repeatability and stable maneuverability during cross-media process while still allowing fly at high speed, such HAUV system can better meet the requirements of algorithm tests and practical applications at sea. The dynamic surface control method with disturbance observer has been developed in recent research to tackle the challenge of HAUVs operate under complex sea states with wind and wave interferences, could be great such algorithms to be testified in the wild ocean to facilitate future applications. Overall, we should not be limited to the constraints of existing research, and combine more specific application scenarios, boldly propose design forms, carefully carry out program demonstration, actively carry out experimental analysis, and strive to enrich the theoretical system so that the academic community and the industry can quickly form a beneficial technical accumulation with breadth and depth in the development of amphibious vehicles. The work of this paper is also a new exploration of the technical forms and critical technical issues of HAUVs. Nevertheless, much work remains to be done.

- **Optimization of water-air two-phase hydrodynamic characteristics of multimodal amphibious vehicles.** Recent work focuses on the concept, principle, and feasibility of the multi-mode amphibious vehicle, but the theoretical and methodological research on the system configuration optimization has not yet been carried out. Therefore, to develop a multi-mode amphibious vehicle that truly meets the requirements of practical applications, the water-air two-phase hydrodynamic characteristics of the system should be analyzed and optimized to design an optimal configuration.
- **Research on the hydrodynamic changes of multimodal amphibious vehicles during the trans-media process.** The hydrodynamic variation of the amphibious vehicle in the cross-domain process is of great value and significance to its structural design and optimization, safety analysis and evaluation, cross-domain performance analysis and evaluation, cross-domain motion modeling, water entry and exit control strategy development and controller design, and extra. Therefore, future research should be combined with the design configuration of the multi-mode amphibious vehicle to carry out in-depth research on its time-varying hydrodynamic characteristics in the cross-domain. The experimental tests, numerical simulations, and theoretical analyses of the time-varying

hydrodynamic characteristics of the multimodal amphibious vehicle should be carried out in conjunction with the design configuration.

- **Optimization improvement of cross-domain motion control algorithms for multi-mode amphibious vehicles.** Recent research on the cross-domain motion control method is still at the stage of theory and simulation testing. In future research, the cross-domain motion control algorithm should be applied to the solid platform of the multi-mode amphibious vehicle to carry out experimental verification of cross-domain motion control under natural conditions. In addition, although the control strategy has good adaptive capability in the face of environmental changes, its application presupposes that the system needs to maintain small-angle operations. The controller design method, which directly unifies the 6DOF motion model of the vehicle, can be improved to achieve adaptive control under arbitrary attitude operation. The control strategy will also be affected by the actuator real output limitation problem, and the introduction of the anti-windup solution can be considered to make the controller better able to cope with the actuator saturation phenomenon.
- **Study of thruster operating performance variation and water impact during cross-domain.** In recent research, when studying the cross-domain motion control problem, the effect of medium change on the thruster operating characteristics is commonly simplified, and the thruster output change is considered a simple linear relationship with density change. However, the actual interaction of the thruster with the water surface during the trans-media process and the problem of the ground effect generated near the water surface are both too complex to be considered. Therefore, the theoretical analysis, numerical simulation, and experimental methods should be combined to investigate the change of thruster performance during the trans-media process in depth.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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